Project Synopsis Template

1. Telugu Movie Recommendation System

Team Details

Student Name	College Name	Branch	Mobile No	Mail ID	Role (Lead/Member)
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Program Track and Domain

Program Track (Eng-HUB / Eng-Spoke / Science-HUB / Science-Spoke / Commerce-Spoke)	Domain (Analytics / Machine Learning / Deep Learning / GenAI)
Eng-HUB	Machine Learning

Guide Details

Mr. MANOJ KUMAR

2. Problem Statement

With the rapid growth of Telugu cinema (Tollywood), thousands of movies are available across different OTT platforms and websites. Users often face difficulty in selecting movies based on their preferences such as genre, actors, ratings, or past watching history. Most global recommendation

systems focus mainly on Hollywood or Bollywood content, while Telugu-specific personalized recommendation systems are limited.

This project aims to build a machine learning-based Telugu Movie Recommendation System that suggests movies based on user preferences and similarity analysis. The system will reduce user effort in searching and improve user experience by providing personalized movie suggestions.

3. Objectives

- To collect and preprocess Telugu movie dataset.
- To analyze movie features such as genre, director, cast, ratings, and year.
- To implement content-based and collaborative filtering algorithms.
- To build a personalized movie recommendation engine.
- To evaluate model performance using similarity metrics.
- To deploy the system using Streamlit as a web application.

4. Methodology / Implementation Plan

Students provide a simple overview of how the project will be carried out.

Data Collection

- Collect Telugu movie dataset (movie name, genre, rating, cast, director, etc.)

Data Preprocessing

- Handle missing values
- Remove duplicates
- Feature engineering
- Convert text data into numerical format (TF-IDF / Count Vectorizer)

Exploratory Data Analysis (EDA)

- Genre distribution
- Rating analysis

Model Building

- Content-Based Filtering (Cosine Similarity)
- Collaborative Filtering
- Hybrid Recommendation Approach
- Model Evaluation

- Precision
- Recall
- Similarity score validation

Deployment

- Build interactive UI using Streamlit
- Deploy locally or on cloud

Algorithms / Models Used

- Content-Based Filtering
- Collaborative Filtering
- Cosine Similarity
- K-Nearest Neighbors (KNN)
- TF-IDF Vectorization

Tools & Technologies

- Python
- Pandas, NumPy
- Scikit-learn
- Matplotlib / Seaborn
- Jupyter Notebook
- Streamlit
- VS Code

5. Resources and Timeline

A. Hardware & Software Requirements

Hardware

- Laptop with minimum 8GB RAM
- Intel i5 or above processor

Software

- OS: Windows / Linux / Mac
- Python 3.x
- Jupyter Notebook
- VS Code
- Streamlit

B. Dataset
Dataset available: Yes

Source: Kaggle / IMDb Telugu dataset / Manually collected dataset

Fields: Movie Name, Genre, Director, Cast, Rating, Year

If dataset is not fully available, data will be collected manually from public movie databases.

C. Project Timeline

Phase	**Task**	Tentative Completion Date
Phase 1	Dataset Collection	Week 1
Phase 2	Data Cleaning & Preprocessing	Week 2
Phase 3	EDA & Visualization	Week 3
Phase 4	Model Development	Week 4
Phase 5	Model Evaluation	Week 5
Phase 6	Deployment using streamlit	Week 6
Phase 7	Final documentation	Week 7

6. Expected Outcomes / Conclusion

By the end of the project:

- A fully functional Telugu Movie Recommendation System will be developed.
- Users will receive personalized movie suggestions.
- The system will improve movie selection efficiency.
- A Streamlit-based web application prototype will be created.
- The project will demonstrate practical implementation of machine learning algorithms.

7. References

- Kaggle Movie Datasets
- Scikit-learn Documentation
- Streamlit Official Documentation
- Research Papers on Recommendation Systems
- YouTube Tutorials on Movie Recommendation Systems